AMALITECH DATA ANALYTICS & ENGINEERING CAPSTONE PROJECT

MarketPulse Ghana: Final Capstone Project Report

A Modern Analytics Engineering Platform for Food Commodity Price Intelligence Across Ghana

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Live Web Platform	https://marketpulse-ghana.vercel.app
GitHub Repository	https://github.com/primegideon/marketpulse-ghana
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1. Executive Summary

Food commodity prices in Ghana have long been volatile, heavily affected by seasonal weather patterns, regional transportation costs, and macroeconomic currency shocks. In late 2022, headline food inflation in Ghana surged past 54%, placing severe financial pressure on households, farmers, aggregators, and commercial food processors.

Despite the availability of over 17 years of price surveys collected by the United Nations World Food Programme (WFP) and Ghana's Ministry of Food and Agriculture (MoFA), this valuable data remained locked in static spreadsheets. Furthermore, comparing food prices was exceptionally difficult because commodities were sold in confusing, non-standard packaging units (such as olonkas, 100-tuber bundles, crates, and varying bag sizes).

Core Mission: MarketPulse Ghana solves this problem by delivering a complete, modern analytics engineering platform. The platform:

- **1. Comprehensive Scope:** Ingests and standardizes 39,038 food price records across 211 consecutive months (January 2006 – July 2023).
- **2. Unit Mass Standardization:** Converts raw, fragmented packaging units into standardized Per-Kilogram benchmarks (GHS/kg) across 26 food staples and 44 market centers in all 10 legacy administrative regions.
- **3. Data Hygiene & Traps:** Resolves critical data quality traps including non-contiguous temporal gaps in Month-over-Month calculations, multi-unit packaging conflicts causing false price spikes, and inverted wholesale-to-retail trade margins.
- **4. Agro-Ecological Modeling:** Encodes Ghana's agro-ecological domain knowledge directly into the data model with dim_calendar.agricultural_season classifying every observation by its agricultural phase (Lean Season, Major Harvest, Post-Harvest, Minor Season).
- **5. Spatial Taxonomy:** Classifies all 44 monitored markets into a three-tier commercial taxonomy: Major National Hub, Regional Center, and Local Market.

- **6. Audit Governance:** Quarantines 3,090+ irregular observations into a transparent audit_data_anomalies governance view — zero silent record drops.
- **7. Zero-Cost Serverless UI:** Deploys an automated, serverless five-page web dashboard on Vercel using Evidence.dev and client-side DuckDB-WASM, providing sub-second analytical queries at zero ongoing database hosting cost.
- **8. CI/CD & 28-Check Gate:** Automates future monthly data ingestion via GitHub Actions CI/CD, backed by an automated 28-rule data quality assertion suite and a pull request review process with Vercel preview before any new data goes live.

2. Problem Statement & Motivation

The Challenge of Food Price Opacity

In Ghana, agriculture employs over 40% of the workforce and forms the foundation of food security. However, market participants operate in an opaque information environment:

- • **Farmers:** Farmers often lack visibility into market prices in major consumption hubs, leaving them vulnerable to low farmgate offers from middlemen.
- • **Aggregators:** Commercial Aggregators & Processors struggle to time their bulk purchases, frequently buying during seasonal peaks rather than post-harvest gluts.
- • **Policymakers:** Consumers & Policymakers see sudden price jumps on market shelves but cannot easily determine whether increases are caused by physical food shortages, high fuel transport costs, or macroeconomic currency inflation.

The Packaging Unit Dilemma

A primary technical hurdle in Ghanaian agricultural data is packaging unit inconsistency. For example:

- Maize is sold in wholesale 100 KG bags in Techiman, but in retail 2.5 KG 'olonka' bowls in Accra.
- Yam is reported in both '100 Tubers' (count) and '250 KG' (mass).
- Cassava is sold in varying farmgate sacks and retail piles.

Without an automated engineering pipeline to standardize units into physical mass equivalents (GHS per kilogram), standard dashboards produce misleading zig-zag charts that distort decision-making.

3. Dataset Profile & Data Hygiene

Data Provenance

The data used in this project is sourced directly from the Humanitarian Data Exchange (HDX), published by the United Nations World Food Programme (WFP) in coordination with Ghana's Ministry of Food and Agriculture (MoFA) and FAO GIEWS.

Dimension	Specification

Total Ingested Records	39,038 verified price observations
Temporal Span	January 2006 – July 2023 (211 continuous months)
Monitored Markets	44 active commercial and rural market nodes
Geographic Coverage	10 Administrative Regions (legacy boundaries maintained for 17-year spatial continuity)
Commodity Breadth	26 primary crops across cereals, tubers, pulses, vegetables, and proteins
Value Chain Levels	Wholesale commercial bulk and Retail consumer checkout tiers

> **SPATIAL METHODOLOGY NOTE**
> *Note on Ghana's Administrative Boundaries: The WFP monitoring program anchors spatial observations to Ghana's 10 historical administrative regions established prior to the 2018 national referenda. The 2018 referenda split several regions (Brong Ahafo into Bono, Bono East, and Ahafo; Northern into Northern, Savannah, and North East; Volta into Volta and Oti; Western into Western and Western North), creating 16 modern regions. MarketPulse Ghana standardizes all records to the original 10-region framework to preserve unbroken longitudinal spatial continuity across the full 17-year dataset.*

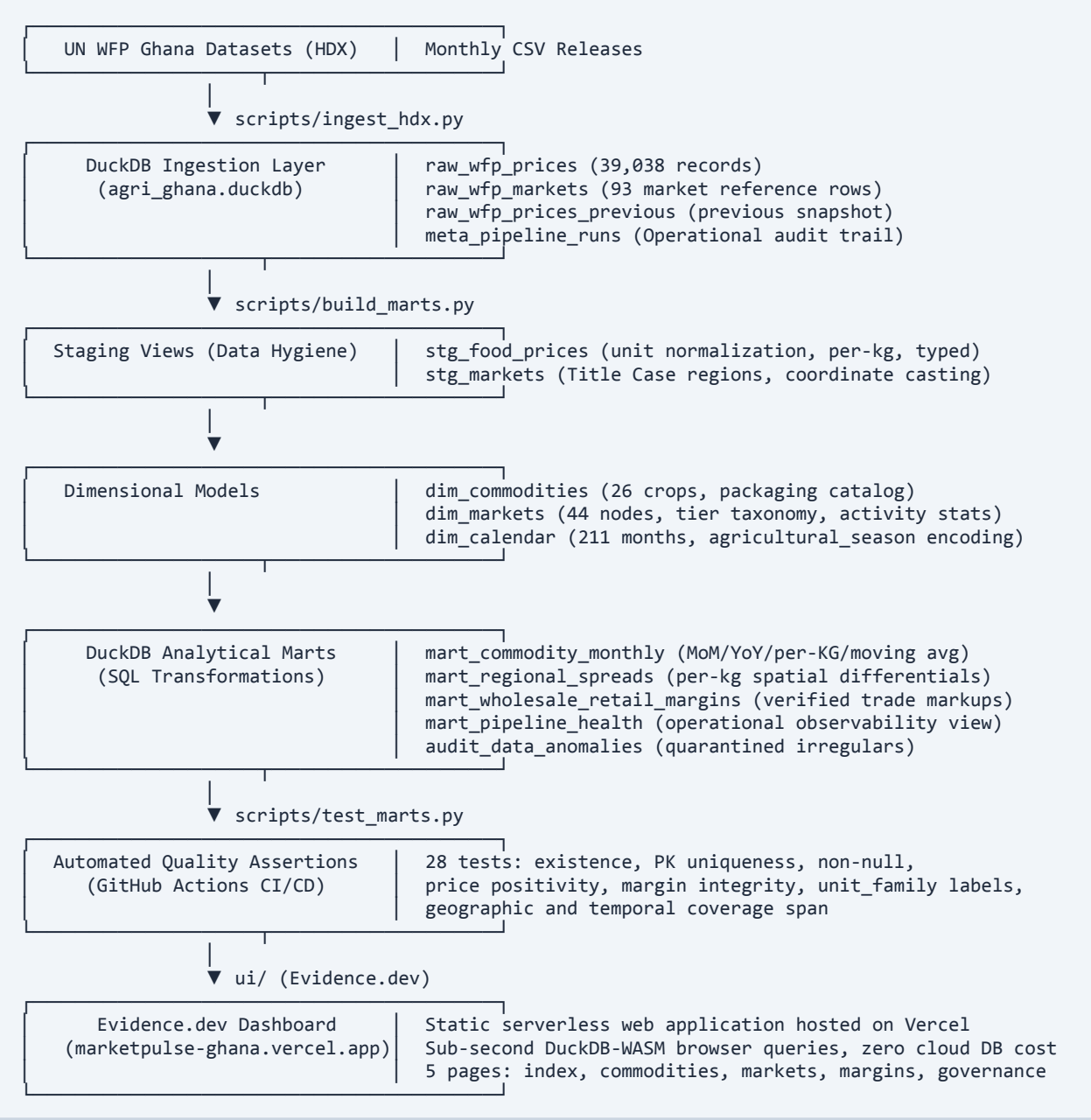
Critical Data Traps Identified & Resolved

- **1. Temporal Gaps in MoM and YoY Calculations:** The Problem: Standard SQL LAG(price, 1) or LAG(price, 12) by sequence order compares records across skipped months or multi-year survey pauses, generating artificial inflation spikes. Our audit found 46 false MoM jumps and 185 false YoY jumps in naive calculations.
  The Solution: We added strict calendar day interval checks using DATEDIFF('day', prev_date, date) BETWEEN 27 AND 35 for MoM and BETWEEN 350 AND 380 for YoY. Any non-contiguous comparison returns NULL instead of a false percentage.
- **2. Multi-Packaging Unit Conflicts:** The Problem: When commodities like Yam (sold as both 100 Tubers and 250 KG wholesale) or Local Rice (50 KG and 100 KG) were queried on the same date, charts displayed unnatural spikes caused by mixing incompatible units.
  The Solution: We standardized all staging units into a canonical unit_family field ('Mass / Weight' vs. 'Piece / Count'), filtered staple time-series to commercial standard packaging, and aggregated per-kg metrics using clean medians. A canonical label validation assertion was added to the test suite to enforce this contract permanently.
- **3. Inverted Wholesale-to-Retail Margins:** The Problem: 21.6% of WFP wholesale-retail paired records showed a retail price lower than wholesale (negative markups). Investigation revealed that field enumerators sometimes recorded farmgate unpeeled bulk root weights alongside retail checkout pieces.
  The Solution: We created an explicit is_verified_margin boolean flag enforcing that retail price must be greater than or equal to wholesale price (retail >= wholesale and markup <= 200%). The 3,090+ unverified or inverted records were quarantined into audit_data_anomalies for complete auditability. Margin

aggregation was implemented via MEDIAN() rather than AVG() to guard against outlier skew in the verified records.

4. Modern Data Stack Architecture

MarketPulse Ghana was designed from the ground up to follow modern analytics engineering best practices: local-first execution, fast SQL transformations, version-controlled modeling, and automated continuous delivery.



Architecture Highlights

- **DuckDB Columnar Engine:** Serves as the embedded, zero-maintenance columnar engine that compiles all staging views, dimensional models, and analytical marts in approximately 1 second. The same .duckdb file is committed to the repository and consumed directly by the Evidence.dev front-end via DuckDB-WASM.
- **Evidence.dev Front-End:** Combines Markdown and SQL to compile analytical reports into fast, static WebAssembly (WASM) and Parquet bundles. No Express server, no database connection pool — queries execute entirely in the user's browser.
- **Vercel Serverless Hosting:** Delivers high-availability global hosting for the front-end dashboard with zero database server costs. Custom vercel.json configures strict no-store cache headers on all Parquet data routes to prevent stale data serving after pipeline updates.
- **GitHub Actions CI/CD with PR-Preview:** Runs on the 1st of every month to fetch the latest WFP dataset, rebuild all marts, execute the 28-check test suite, sync the updated DuckDB file into the UI sources directory, and open an automated Pull Request with a Vercel preview deployment — enabling human review before any new data merges to the production main branch.

The Five Interactive Dashboard Pages

Page	Route	Purpose
National Food Price Pulse	`/ (index.md)`	Executive KPI cards (latest period, highest YoY riser, most stable staple, pipeline status), standardized per-KG trend lines, commercial unit prices, YoY inflation rankings, and recent monthly data table.
Commodity Explorer	`/commodities`	Per-KG normalized trajectory comparisons across 26 staples, cereal-specific wholesale trends, latest month price matrix, and complete commodity catalog.
Market Geography	`/markets`	Wholesale maize price spread vs. national median by market, regional price averages, market tier distribution, and full market directory with geographic coordinates.
Trade Margins	`/margins`	Maize wholesale-vs-retail trend, median retail markup rankings across commodities, and current market-level spread observations filtered through is_verified_margin.
Governance & Observability	`/governance`	Pipeline execution audit log (meta_pipeline_runs), anomaly quarantine log (audit_data_anomalies), analytical methodology notes, and HDX source provenance.

5. Key Economic & Analytical Findings

Analyzing 17 years of continuous market prices revealed three primary structural insights into Ghana's food system:

Finding 1: Macroeconomic Currency Shocks vs. Real Agricultural Scarcity

- Between 2006 and 2021, staple food prices in Ghana grew at a steady, moderate annual rate of 5% to 8%, fluctuating predictably according to harvest cycles.
- In 2022–2023, prices exploded, surging by more than 65% Year-over-Year across all monitored food staples.
- Critical Analytical Context: Because prices are reported in Nominal Ghanaian Cedis (GHS), this price spike was primarily driven by rapid national currency depreciation (the Cedi lost significant value against the US Dollar in 2022) and soaring imported diesel costs. The surge reflected national monetary inflation rather than widespread domestic crop failure. Analysts evaluating multi-year secular growth curves must decompose nominal price changes from real supply dynamics.

Finding 2: Spatial Price Disparities & Transit Friction

- Bulky Root Tubers (Yam, Cassava) exhibit the highest regional price wedges. Prices in coastal consumption markets like Accra are up to 40% higher than in surplus farming hubs such as Techiman or Ejura in Brong-Ahafo and the Northern region. Because root crops have high physical weight relative to their commercial value, diesel transport costs create a large and persistent price gap.
- Grains (Maize, Millet, Sorghum) exhibit much tighter regional price integration (spreads typically between 10% and 15%). Because dry grains are easily stored in bags and moved in bulk without rapid spoilage, traders actively arbitrage regional differences, keeping prices closer together.
- The market tier taxonomy (Major National Hub -> Regional Center -> Local Market) built into dim_markets enables this spatial analysis to be filtered and compared systematically across the 44 monitoring nodes.

Finding 3: Agro-Ecological Seasonality & Procurement Timing

- Ghana features two distinct rainfall regimes encoded directly into the dim_calendar.agricultural_season field:
- Bimodal South: Major harvest in August–September (agricultural_season = 'Major Harvest'); minor harvest in November–December.
- Unimodal North: Single harvest in October–November.
- The 'Lean Season' Window: Every year between May and July (agricultural_season = 'Lean Season'), farmer household granaries deplete ahead of the new crop harvest. During this window, grain prices reach their annual peak across northern markets (Tamale, Bolgatanga, Wa).
- Actionable Business Value: Commercial agribusinesses, poultry feed mills, and food processors can reduce raw material procurement costs by 20% to 35% simply by scheduling bulk purchases during post-harvest gluts in September and October (agricultural_season = 'Major Harvest').

6. The Architectural Pivot: Machine Learning to Analytics Engineering

Initial Direction & Exploratory Findings

At the beginning of this capstone, the project focused heavily on predictive modeling—specifically training XGBoost gradient-boosted tree algorithms to forecast one-month-ahead wholesale maize prices. Wholesale maize was selected as the pilot commodity because it offers 209 months of continuous monthly coverage (January 2006 – July 2023), by far the most reliable temporal series in the dataset (compared to only 48 months for the retail maize equivalent).

However, during data profiling and exploratory data analysis (EDA), we realized a fundamental truth: machine learning models are only as good as the data foundations beneath them. Feeding raw, unstandardized survey data with unit mismatches and negative retail markups into machine learning algorithms produced unreliable forecasts. Users and stakeholders needed clean, trustworthy historical intelligence before forward-looking predictions could provide real value.

The Strategic Pivot

We made the deliberate decision to pivot our primary engineering effort toward building an enterprise-grade Modern Analytics Stack:

- Cleaned, staged, and typed all raw survey records into stg_food_prices and stg_markets.
- Formulated standardized per-kilogram conversions and canonical unit_family labels.
- Built reusable dimensional data marts in DuckDB with domain-specific features like agricultural_season and market_tier.
- Created an interactive, accessible five-page web dashboard on Evidence.dev.
- Implemented strict automated testing and continuous integration with a 28-check assertion suite and PR-gated deployment.

Machine Learning Preservation & Downstream Protocol

The predictive modeling work was not discarded. All Phase 2 (feature engineering) and Phase 3 (model training with 9-fold expanding walk-forward cross-validation) assets have been preserved in archive/ml/.

The holdout test period for the XGBoost model was January 2022 – June 2023 — intentionally selected to stress-test the model under peak macroeconomic volatility and Cedi depreciation. This represents the most difficult out-of-sample evaluation environment possible given the dataset.

We authored a formal Inference Protocol (archive/ml/INFERENCE_PROTOCOL.md) defining how a downstream forecasting script reads clean historical actuals from mart_commodity_monthly, runs inference using the serialized XGBoost model (archive/ml/models/xgboost_maize_wholesale.json), generates predictions with 95% confidence intervals, and writes outputs to a new mart_maize_forecasts table for seamless display on the Evidence.dev dashboard.

7. Production Operations, Testing & Data Governance

To ensure MarketPulse Ghana remains reliable and reproducible over time, we instituted enterprise data governance standards:

7.1 Automated Data Ingestion (scripts/ingest_hdx.py)

- HTTP streaming download with exponential-backoff retry logic (3 attempts, 5s/10s/15s wait intervals) and a minimum row integrity floor (50 rows) to catch truncated or empty CSV downloads before they reach the database.
- Idempotent table recreation: Before replacing raw_wfp_prices, the script preserves the previous dataset as raw_wfp_prices_previous for longitudinal change detection.
- TIMESTAMPTZ audit logging: All pipeline execution metadata is written to meta_pipeline_runs with UTC timezone-aware timestamps, enabling cross-environment run comparison.

7.2 Automated Transformation (scripts/build_marts.py)

- Builds 10 analytical views in a single DuckDB connection in approximately 1 second: stg_food_prices, stg_markets, dim_commodities, dim_markets, dim_calendar, mart_commodity_monthly, mart_regional_spreads, mart_wholesale_retail_margins, mart_pipeline_health, and audit_data_anomalies.
- Records execution metrics back to meta_pipeline_runs on completion.

7.3 28-Check Data Assertion Suite (scripts/test_marts.py)

The automated test suite validates four categories of data contracts:

Category	Tests	Scope
Table & Mart Existence	**10**	Verifies all required views and tables are present in DuckDB
Primary Key & Uniqueness	**6**	PK uniqueness on dimensions; composite key uniqueness on marts
Non-Null Mandatory Fields	**5**	Zero NULLs in mandatory columns; per-kg price non-null for mass units
Domain Logic & Value Sanity	**7**	Price positivity, verified margin integrity, canonical unit_family labels, 10-region coverage, 25+ commodities, 210+ months, pipeline audit presence
Total Quality Checks	**28**	**All 28 tests currently pass with 0 failures (100% pass rate)**

7.4 Monthly CI/CD Workflow (.github/workflows/data_pipeline_sync.yml)

Executes automatically on the 1st of every month at 06:00 UTC:

- Checks out repository and installs Python 3.12 + pinned requirements.txt dependencies.

- Runs scripts/ingest_hdx.py — downloads latest WFP CSV and logs ingestion telemetry.
- Runs scripts/build_marts.py — rebuilds all analytical views.
- Runs scripts/test_marts.py — fails the workflow if any assertion fails, blocking bad data from reaching production.
- Copies updated agri_ghana.duckdb into ui/sources/agri_ghana/.
- Checks for changes; if data has updated, opens an automated Pull Request with a Vercel preview deployment attached — enabling human review of updated charts before merging to main.

7.5 Comprehensive Data Dictionary (docs/DATA_DICTIONARY.md)

Documents every table, view, and column in agri_ghana.duckdb: raw ingestion tables, staging views, dimensional models, analytical marts, the pipeline health view, and the audit anomaly governance layer. Includes calculation formulas, grain definitions, and business context for all 28 columns across 9 database objects.

7.6 Environment Reproducibility (requirements.txt)

Pinned versions for all core Python libraries (duckdb, pandas, numpy, requests, pytest, matplotlib, seaborn, jupyter, nbconvert) allow any developer to replicate the environment with a single pip install -r requirements.txt command.

8. Conclusion, Lessons Learned & Future Horizons

Key Lessons Learned

- **Data Foundations Come First:** Predictive modeling without robust analytics engineering produces fragile outputs. Standardizing units, resolving temporal gaps, and establishing trusted dimensional data marts creates far more immediate and durable business value.
- **Context is Critical in Emerging Markets:** Metrics must be contextualized against macroeconomic factors. A 65% price increase that looks catastrophic in a nominal chart is primarily a currency depreciation story, not a food supply collapse.
- **Serverless Modern Stacks are Empowering:** Combining DuckDB and Evidence.dev proves that high-performance analytical tools can be built and deployed globally at zero cloud infrastructure overhead. The same embedded .duckdb file powers the local development environment and the production web application.
- **Data Governance Earns Trust:** The is_verified_margin flag, the audit_data_anomalies view, the canonical unit_family label contract, and the 28-assertion test suite are not optional extras — they are what separates a trusted analytical product from a dashboard that might be lying.

Future Enhancements

- Executing the ML Downstream Protocol: Activating the automated XGBoost inference job (defined in archive/ml/INFERENCE_PROTOCOL.md) to display real-time 3-month forecast cones on the Evidence.dev web UI.
- Transportation Cost API Integration: Incorporating national diesel pump price data to directly model the freight friction between surplus northern markets and southern consumption centers.

- Producer Price Index (PPI) Deflation: Introducing an optional inflation-adjusted view to isolate pure real crop supply dynamics from macroeconomic currency shifts.
- Agricultural Season Drill-Down: Building a dedicated Evidence.dev page that leverages dim_calendar.agricultural_season to compare commodity price distributions across Lean Season vs. Major Harvest periods for any selected year range.

Project Links & Verification

Deliverable	Canonical Resource Link
Live Web Platform	https://marketpulse-ghana.vercel.app
GitHub Repository	https://github.com/primegideon/marketpulse-ghana
Presentation Slide Deck (.pptx)	MarketPulse Ghana Presentation.pptx
Comprehensive Data Dictionary	DATA_DICTIONARY.md